

- Description

This module manages a smart conveyor belt system by continuously monitoring the load weight on the belt (in kg) and adjusting the belt speed (in m/s) accordingly.

1. If the load weight exceeds a predefined threshold, the belt speed is reduced (e.g., decreased by 20%) to prevent overloading.
2. If the load weight is within the safe limit, the belt maintains its optimal speed.

This system helps maintain optimal throughput and ensures the safe operation of the conveyor belt in a production environment.

- Specification

module SmartConveyorBeltControl

var currentLoad, loadThreshold, optimalSpeed, currentSpeed: real

process updateLoad(newLoad: real) returns (currentLoad: real, currentSpeed: real)

functional scenario form:

currentLoad > loadThreshold && currentSpeed = optimalSpeed * 0.8, output "ALERT: Load (" + currentLoad + " kg) exceeds threshold (" + loadThreshold + " kg). Speed reduced to " + currentSpeed + " m/s.", Conveyor belt speed adjusted to reduced value.

||

currentLoad <= loadThreshold && currentSpeed = optimalSpeed, output "Load (" + currentLoad + " kg) is within safe limits. Speed maintained at " + currentSpeed + " m/s.", Conveyor belt speed remains optimal.

end process

end module

- Java Code

```
package code;
public class c99_SmartConveyorBelt {
    private double loadThreshold;
    private double optimalSpeed;
    private double currentLoad;
    private double currentSpeed;
    public c99_SmartConveyorBelt(double loadThreshold, double optimalSpeed) {
        this.loadThreshold = loadThreshold;
        this.optimalSpeed = optimalSpeed;
        this.currentLoad = 0.0;
        this.currentSpeed = optimalSpeed;
    }
    public void updateLoad(double newLoad) {
        currentLoad = newLoad;
        adjustBeltSpeed();
    }
}
```

```

        System.out.printf("Current Load: %.2f kg, Conveyor Belt Speed:
%.2f m/s%n",
                        currentLoad, currentSpeed);
        if (currentLoad > loadThreshold) {
            System.out.println("ALERT: Load exceeds threshold. Conveyor
belt speed reduced.");
        } else {
            System.out.println("Load is within safe limits. Conveyor be
lt operating at optimal speed.");
        }
    }
    private void adjustBeltSpeed() {
        if (currentLoad > loadThreshold) {
            currentSpeed = optimalSpeed * 0.8;
        } else {
            currentSpeed = optimalSpeed;
        }
    }
    public double getCurrentSpeed() {
        return currentSpeed;
    }
    public static void main(String[] args) {
        c99_SmartConveyorBelt belt = new c99_SmartConveyorBelt(100.0, 2.0)
;
        // testcase-VT:
        //     belt.updateLoad(80.0);
        //     belt.updateLoad(120.0);
        //     belt.updateLoad(95.0);

        // testcase-FT:
        //     belt.updateLoad(-1.2007939425236886E308);
        //     belt.updateLoad(-1.1867492506025853E308);
        //     belt.updateLoad(-1.7836486571938841E308);

        // testcase-Z3:
        //     belt.updateLoad(80.0);
        //     belt.updateLoad(120.0);
        //     belt.updateLoad(95.0);

        // testcase-uVT:
        belt.updateLoad(80.0);
        belt.updateLoad(120.0);
        belt.updateLoad(95.0);
    }
}

```

```
}
```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c99_SmartConveyorBelt.java	100.0 %	4	0	4
▼ c99_SmartConveyorBelt	100.0 %	4	0	4
● updateLoad(double)	100.0 %	2	0	2
■ adjustBeltSpeed()	100.0 %	2	0	2
● c99_SmartConveyorBelt(double,		0	0	0
\$ main(String[])		0	0	0

3. Vibration testing instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	21,714	21,800
▼ src	0.4 %	86	21,714	21,800
▼ code	0.4 %	86	21,714	21,800
▼ c99_SmartConveyorBelt.java	96.6 %	86	3	89
▼ c99_SmartConveyorBelt	96.6 %	86	3	89
● updateLoad(double)	100.0 %	37	0	37
■ adjustBeltSpeed()	100.0 %	18	0	18
● c99_SmartConveyorBelt(double,	100.0 %	15	0	15
\$ main(String[])	100.0 %	16	0	16

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c99_SmartConveyorBelt.java	50.0 %	2	2	4
▼ c99_SmartConveyorBelt	50.0 %	2	2	4
● updateLoad(double)	50.0 %	1	1	2
■ adjustBeltSpeed()	50.0 %	1	1	2
● c99_SmartConveyorBelt(double,		0	0	0
\$ main(String[])		0	0	0

5. Fuzz testing instruction coverage: 84.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	75	21,725	21,800
▼ src	0.3 %	75	21,725	21,800
▼ code	0.3 %	75	21,725	21,800
▼ c99_SmartConveyorBelt.java	84.3 %	75	14	89
▼ c99_SmartConveyorBelt	84.3 %	75	14	89
● updateLoad(double)	89.2 %	33	4	37
■ adjustBeltSpeed()	61.1 %	11	7	18
● c99_SmartConveyorBelt(double,	100.0 %	15	0	15
\$ main(String[])	100.0 %	16	0	16

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c99_SmartConveyorBelt.java	100.0 %	4	0	4
▼ c99_SmartConveyorBelt	100.0 %	4	0	4
● updateLoad(double)	100.0 %	2	0	2
■ adjustBeltSpeed()	100.0 %	2	0	2
● c99_SmartConveyorBelt(double,		0	0	0
\$ main(String[])		0	0	0

7. Z3 instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	21,714	21,800
▼ src	0.4 %	86	21,714	21,800
▼ code	0.4 %	86	21,714	21,800
▼ c99_SmartConveyorBelt.java	96.6 %	86	3	89
▼ c99_SmartConveyorBelt	96.6 %	86	3	89
● updateLoad(double)	100.0 %	37	0	37
■ adjustBeltSpeed()	100.0 %	18	0	18
● c99_SmartConveyorBelt(double,	100.0 %	15	0	15
● main(String[])	100.0 %	16	0	16

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c99_SmartConveyorBelt.java	100.0 %	4	0	4
▼ c99_SmartConveyorBelt	100.0 %	4	0	4
● updateLoad(double)	100.0 %	2	0	2
■ adjustBeltSpeed()	100.0 %	2	0	2
● c99_SmartConveyorBelt(double,		0	0	0
● main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	21,714	21,800
▼ src	0.4 %	86	21,714	21,800
▼ code	0.4 %	86	21,714	21,800
▼ c99_SmartConveyorBelt.java	96.6 %	86	3	89
▼ c99_SmartConveyorBelt	96.6 %	86	3	89
● updateLoad(double)	100.0 %	37	0	37
■ adjustBeltSpeed()	100.0 %	18	0	18
● c99_SmartConveyorBelt(double,	100.0 %	15	0	15
● main(String[])	100.0 %	16	0	16